
Profile

MIT 2025 B.S. and M.Eng. graduate and U.S. citizen specializing in **backend systems** and **API architecture**, with experience building **scalable cloud services** in AWS.

Education

Massachusetts Institute of Technology (MIT)	Cambridge, MA
M.Eng. in Computer Science	GPA: 5.0 / 5.0
Computer Networks; Systems Security; Distributed Systems; Robotics	
B.S. in Computer Science	GPA: 4.7 / 5.0
Software Performance Engineering; Computational Architecture; Design of Algorithms; Systems Engineering; Network Security; Machine Learning	

Experience

Software Engineer	Capital One	2025 - Current
- Building backend connecting customers and vendors via multiple payment strategies, improving customer experience. - Developed REST APIs for customer data validation, automating verification and reducing transaction errors. - Deployed cloud infrastructure using AWS Lambda, Route 53, and DynamoDB to enhance scalability and reliability.		
Back-end Software Engineer, Intern	Capital One	Summer 2023 / 2024
- Developed Spring Boot API to generate custom configs, resulting in dynamic web pages. - Enhanced back-end security by implementing input validation layer. - Implemented and deployed backend request logging, reducing resolution time by 30%. - Designed PostgreSQL database schema for efficient data storage.		
AI Software Developer	MIT Media Lab	Spring 2024
- Boosted data storage efficiency by 30% using SQL. - Designed API architecture, streamlining feature implementation through a clear roadmap. - Automated context retrieval for the model by implementing a RAG parser.		

Projects

Raft Distributed System	Go / RPC / Parallelization
- Implemented fault tolerant Raft Distributed System, using RPC and Go. - Optimized worker and leader algorithms through parallelized RPC calls. - Thoroughly tested the implementation using Go's robust testing framework.	
Ray Tracer + Multi-Body Simulator	C / C++ / AWS / Multi-Core
- Utilized OpenCilk for multi-core processing. - Optimized algorithms based on Span and Work of paralleled code. - Utilized AWS for better performance testing.	
StarBattle Video Game	Typescript / Full-Stack
- Implemented feature where the application handles concurrent inputs from users. - Created Python back end server to retrieve and store games.	

Tools

- Languages:** Python; Java; Typescript; Go; C;
- Libraries:** OpenCilk; Springboot; Fast API; Maven; Gradle
- Stacks:** Git; Lambda; PostgreSQL;